

Yaotian Zhang

Github: github.com/dandelionskyy

Email: yaotianzhang94@gmail.com

EDUCATION

Southwest Jiaotong University

Sep 2023 – Jun 2027

B.Eng. in Electronic Information Engineering

- GPA: 3.68/4.0
- Overall Score: 91.19/100
- Rank: 13/79 (Top 16%)

RESEARCH EXPERIENCE

Contact-Aware SLAM Optimization for Legged Robots

Mar 2025 – Aug 2025

Methods: Fast-LIO2, RANSAC, State Machine

- Investigated localization degradation in legged robots caused by body vibrations during locomotion.
- Developed a contact-state detection method using foot-end sensor feedback and designed a state-machine framework for adaptive mode switching within Fast-LIO2.
- Extracted ground planes using **RANSAC** during stable support phases and estimated surface normal vectors to construct vertical geometric constraints.
- Integrated plane-based constraints into the state estimation process to correct accumulated drift along the z-axis.
- Reduced vertical localization errors during robot motion while maintaining an average processing time below **75 ms**.
- Improved z-axis localization accuracy to approximately **2 cm** on a real quadruped robot platform.

Urban Traffic Risk Prediction and Assessment

May 2024 – Jul 2025

Methods: YOLOv8, DeepSORT, SVM

- Constructed a large-scale traffic interaction dataset containing over **300,000 annotated images** collected from urban signalized intersections.
- Developed a real-time vehicle and pedestrian detection and tracking pipeline based on **YOLOv8** and **DeepSORT**.
- Extracted spatiotemporal interaction features from road users and designed a risk assessment framework for traffic conflict analysis.
- Implemented an **SVM-based** prediction model for hazardous event classification and risk evaluation.
- Proposed a machine-learning-based traffic risk prediction framework integrating object detection, trajectory tracking, and behavior analysis.
- Achieved an object detection performance of **mAP@0.5 = 0.934**, with detection accuracy exceeding **0.90** for vehicle categories and an average precision above **0.50** for Vulnerable Road Users (VRUs).
- Achieved **98% classification accuracy** and an **F1-score of 0.96** for low-, medium-, and high-risk traffic event assessment using the proposed SVM-based risk prediction model.

TECHNICAL SKILLS

Programming Languages: C++, Python, MATLAB, C

Robot Software: ROS1, ROS2, Nav2, Fast-LIO2, VINS-Mono

Robot Perception & Localization: LiDAR-Inertial Odometry, AprilTag, OpenCV, PCL

Machine Learning & Computer Vision: PyTorch, YOLOv8, DeepSORT, SVM

Embedded Platforms: STM32, NVIDIA Jetson Orin Nano

Simulation: Gazebo, MuJoCo

PROJECT EXPERIENCE

Quadruped Obstacle Perception and Localization

Mar 2025 – Aug 2025

Technologies: Depth Camera, IMU, AprilTag, Stanley Controller

- Developed a multi-sensor perception framework for a 8-DoF parallel-leg quadruped robot to enable autonomous traversal of stairs, slopes, walls, and other complex obstacles.
- Designed and implemented a depth camera–IMU fusion pipeline for real-time environmental perception and robot state estimation.
- Developed an AprilTag-based target detection module for 6-DoF target localization and robot pose estimation.
- Designed a coordinate transformation and motion-correction strategy to improve target-approaching accuracy.
- Proposed a segmented target-correction method inspired by the Stanley controller, introducing intermediate waypoints at different distance stages to improve convergence performance.
- Achieved localization accuracy below **1 cm** within **3 m** and below **5 cm** within **8 m**, with localization and correction latency below **10 ms**.
- Successfully deployed and validated the system on an 8-DoF parallel quadruped robot platform.

Quadruped Localization, Mapping, and Navigation

Mar 2025 – Jun 2025

Technologies: Livox Mid360, Fast-LIO2, Fast-LIO-Localization, Nav2

- Developed a LiDAR–inertial localization and navigation framework for 12-DoF quadruped robots using Livox Mid360 and Fast-LIO2.
- Designed a point-cloud preprocessing pipeline based on Statistical Outlier Removal (SOR) filtering to improve map quality and localization robustness.
- Built a map-based relocalization module using Fast-LIO-Localization for high-precision localization in previously mapped environments.
- Integrated the localization system with ROS2 Nav2 to enable global path planning, local obstacle avoidance, and autonomous navigation.
- Constructed a complete localization–mapping–relocalization–navigation pipeline for quadruped robots.
- Achieved map-based localization accuracy within **3 cm** in known environments and successfully validated autonomous navigation in complex obstacle courses.

LEADERSHIP & SERVICE

ROBOCON School Robotics Team - Visual Algorithm Group Leader

Aug 2025 – Aug 2026

- Led perception and localization algorithm development for quadruped robots.
- Coordinated algorithm integration and testing within the robotics team.

HONORS & AWARDS

• National First Prize , 24th ROBOCON Bionic Leg Challenge (Indoor Obstacle Course)	2025
• Honorable Mention (H Prize) , Mathematical Contest in Modeling (MCM/ICM)	2024
• National First Prize , MathorCup Mathematical Modeling Challenge	2024
• Chinese Invention Patent (Granted) : Wing Structure Design for Unmanned Aerial Vehicles	
• Provincial First Prize , China Undergraduate Mathematical Contest in Modeling(CUMCM)	2024

SELECTED OPEN-SOURCE PROJECTS

Quadruped Localization and Navigation Framework

GitHub: github.com/dandelionskyy/fast-llo-nav-obstacle

AprilTag-Based Localization and Correction Framework

GitHub: github.com/dandelionskyy/depth_camera_ws